

Yan Wang

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EDUCATION

Wuhan University, Wuhan, China *Sep. 2023-Present*

BS, Biological Science; GPA: 3.91/4.00; Rank: 1/33

Core course: Cell Biology(95) Molecular Biology(96) Genetics(96) Biochemistry(95)

Yale University, New Haven, CT, USA *Expected Aug. 2026-May. 2027*

Visiting Undergraduate Researcher, Scherzer Lab, Adams Center for Parkinson's Disease Research

Tsinghua University High School, Beijing, China *Sep. 2020-Jun. 2023*

National College Entrance Examination, Biology:100/100

LANGUAGE

- **English: TOEFL iBT 106/120 (Reading 29, Listening 28, Speaking 24, Writing 25)**
- **English-taught major courses in the Hongyi Honors Program at Wuhan University.**
- **Received regular academic training in an English-speaking laboratory environment.**

RESEARCH EXPERIENCE

Medical Research Institute, Wuhan University *Jan. 2025-Present*

Joint lab of Prof. Yanxun Yu and Prof. Youngnam Jin

Project title: Analysis of the functional sites of INS-1 neuropeptide in the alcohol perception neural circuit of C.elegans

- Led an independent *C. elegans* neurobiology project investigating how the insulin-like neuropeptide INS-1 regulates context-dependent ethanol valence under high-salt conditions.
- Generated neuron-specific *ins-1* knockout and rescue strains, identifying ASER, AIY, AIZ, and AIB as key candidate neurons involved in INS-1-dependent high-salt ethanol avoidance.
- Combined behavioral assays, calcium imaging, optogenetics and neuronal ablation to show that loss of *ins-1* converts ethanol avoidance into attraction and alters ASER-AIY circuit activity.
- Revealed, through ASER calcium imaging and ethanol dose-dependent behavioral assays, that INS-1 determines the ethanol response range of ASER: loss of *ins-1* shifts ASER sensitivity rightward, while ASER-specific *ins-1* rescue significantly shifts it back toward wild type.
- Proposed NTR-1 as a candidate noncanonical INS-1 receptor by integrating pathway mutant analysis, *ntr-1* phenotyping, and molecular docking.
- Submitted the project as a first-author Neuroscience 2026 scientific abstract.

Project title: Performance Optimization of the POST-IT Proximity Tagging System

- Built a $\sim 10^6$ PafA mutant library by error-prone PCR and performed nine rounds of yeast-display FACS screening to improve POST-IT labeling efficiency.
- Identified 9 key mutations by NGS and constructed combined PafA variants; the 9mt-L1 mutant showed the strongest labeling activity.
- Optimized labeling conditions from 18 h / 40 μ M substrate to 2 h / 10 μ M, increasing threshold-positive cells from 0.5% to 27.2% (>40-fold improvement).
- Preparing to validate improved PafA variants in small-molecule target identification using dasatinib and Hino-related compounds.

Project title: LOGIC -Lasting Optimized Genetic Input Calculator

- Original proposer and core designer of LOGIC; conceived colony-level biocomputing using orthogonal quorum sensing and spatial AHL diffusion.
- Built and screened quorum-sensing reporter systems to identify orthogonal AHL channels for input, output, and carry signals.
- Constructed high-pass and band-pass response modules to implement colony-level AND, OR, XOR, and cascaded logic gates.
- Designed a serial biological adder using AHL diffusion as inter-colony “wires” and integrated modules into a reusable LOGIC toolbox.
- Developed a light-controlled AiiA reset module and quantified >20,000 fluorescence readouts, contributing to an iGEM Gold Medal.

TECHNICAL SKILLS

- Microbial and model organism handling: *E. coli*, yeast, and *C. elegans*.
- Molecular cloning, protein purification, Western blot, FACS, and routine biochemical assays, mammalian cell culture, etc.
- Microinjection, genetic manipulation, calcium imaging, optogenetic manipulation, and behavioral assays in *C. elegans*.
- C, HTML, CSS, etc. Familiar with basic single-cell RNA-seq data analysis workflows in R.
- Basic structural biology tools (AlphaFold and PyMOL), data analysis, data visualization.

ACADEMIC ACTIVITIES

Tsinghua University, Beijing, China

May. 2025

National Forum on China’s “Top-Tier Talent Program 2.0” in Biology; one of two WHU delegates.

Westlake University, Hangzhou, China

Jul. 2025-Aug. 2025

“Frontiers of Life Science” International Undergraduate Summer School

Peking University / Center for Life Sciences, Beijing, China

Jul. 2026

2026 Summer Training Program in Neuroscience and Cognitive Science

HONORS & AWARDS

iGEM Foundation

Nov. 2025

Gold Medal, International Genetically Engineered Machine Competition (iGEM);

Core Team Member and Project Prototype Designer.

Chinese Society of Blotechnology

Aug. 2025

Gold Prize , Regular Track, SynBio Challenges (an international contest on synthetic biology)

Silver Prize , Synthetic Cell Track, SynBio Challenges

Wuhan University

Liu Daoyu Creative Learning Scholarship (awarded to 24 students university-wide)

Jan. 2026

Hongyi Honors College’s Annual Top Ten Scientific and Technological Advances Award

Oct. 2025

First-Class Outstanding Freshman Scholarship

Sep. 2023

Outstanding Student Scholarship (5%)

2024 & 2025

Outstanding Student (5%)

2024 & 2025

Outstanding Student Leader

2024 & 2025